Koralla Kedhar

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Summary

Aspiring Software Development Engineer and B.Tech student at SRM University with hands-on project experience developing Python-based applications across backend systems, computer vision, and machine learning. Familiar with building API-driven services using FastAPI, interactive applications with Streamlit, and image-processing workflows using OpenCV, NumPy, and Pandas. Brings a foundation in programming, database management, data preprocessing, model development, and Git-based collaboration, with an interest in building reliable, maintainable software and solving practical problems through technology.

Education

SRM University, Kattankulathur — Chennai, Tamil Nadu
Bachelor of Technology (B.Tech)
NRI Junior College — Guntur, Andhra Pradesh | *Intermediate (Class XII)*
NRI Indian Springs School — Guntur, Andhra Pradesh | *Class X*

Skills

Programming Languages: C, C++, Python, Java, JavaScript
Web & Backend: HTML, CSS, Node.js, Flask, FastAPI, REST APIs, WebSockets, MySQL
ML & Data: scikit-learn, OpenCV, Pandas, NumPy, computer vision, data preprocessing, classification
Tools & Concepts: Git, Docker, VS Code, Streamlit, Arduino, IoT systems, database management

Experience

Event Coordinator — University Events
– Coordinated technical and cultural events by organizing logistics, registrations, schedules, and participant communication to support smooth event delivery.
– Worked with faculty members and student volunteers to manage multiple tasks and help ensure a positive participant experience.
Volunteer – Lexposium 2025 — SRM School of Law, SRMIST
– Supported event logistics and attendee coordination for Lexposium 2025, including activities associated with the inauguration of the new SRM School of Law building.
– Participated in discussions on judicial reforms, constitutional morality, AI and cybersecurity, and access to justice.

Projects

CodeMate Terminal | GitHub
– Developed an AI-assisted terminal application with a FastAPI backend intended to combine command execution with contextual developer assistance in a single workflow.
– Implemented application components including authentication, WebSocket-based real-time communication, terminal emulation, file management, and system monitoring.
– Included Docker support to package and run the application consistently, with backend components organized for maintainability.
Solar Panel Crack Segmentation | GitHub | Live Demo
– Built a Streamlit image-analysis interface for solar-panel images using OpenCV, NumPy, and Pillow to process images and highlight candidate crack regions.
– Applied image-processing and contour-based analysis to identify visible regions of interest, then displayed detected areas alongside the source image for visual review.
– Calculated the proportion of image area associated with detected contours; outputs depend on image quality and processing thresholds.
Asphalt Quality Detector | GitHub | Live Demo
– Created a Streamlit application for exploring asphalt-related data and generating quality predictions from structured dataset features.
– Used Pandas to prepare and inspect inputs and a scikit-learn Decision Tree classifier to map nine dataset features to a predicted class.
– Connected dataset exploration and model inference in one workflow so users can review data and obtain model outputs; prediction reliability depends on training data and validation.

Achievements

– Member of the university technical club; participated in coding competitions and technical workshops.
– Volunteered in university community-service events and coordinated peer-learning and study groups.
– Conducted Arduino and IoT demonstrations for school and college students.